APPLIED DATA SCIENCE 2

INTRODUCTION:

The goal of this project was to develop deep learning models capable of predicting the genre of films based on their posters and textual overviews. Through a multi-step process, I first built efficient data processing pipelines to prepare the image and text data, then defined and trained appropriate neural network architectures for each modality. Finally, I evaluated the performance of these models on a held-out test set, analyzing their strengths, weaknesses, and the insights they provide.

In this report, I will critically examine the results of the poster classification model and the overview prediction model, discussing their accuracy, strengths, and limitations. I will provide specific examples to illustrate key findings and highlight areas for potential improvement. The report will conclude with an overall assessment of the project, reflecting on the challenges encountered, lessons learned, and potential future directions.

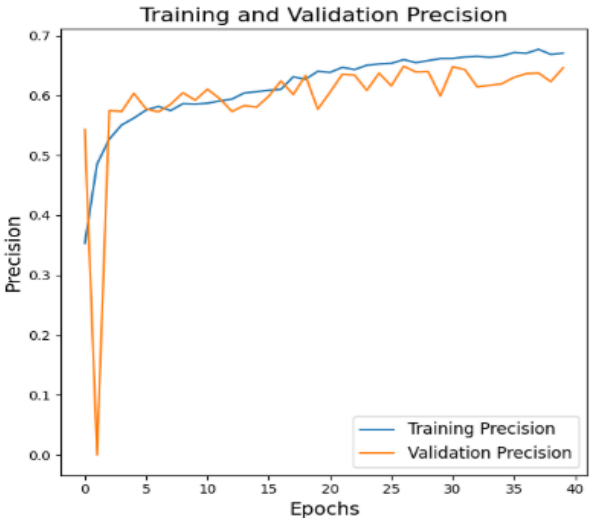
INSIGHTS FROM CNN:

Training and Validation Loss:

- The training loss starts very high and rapidly decreases over the first few epochs.
- The validation loss also starts high but decreases more gradually, remaining higher than the training loss.
- The gap between the training and validation loss narrows over time, indicating the model is generalizing better.
- After around 25 epochs, the training and validation loss curves converge and stabilize.

Training and Validation Precision:

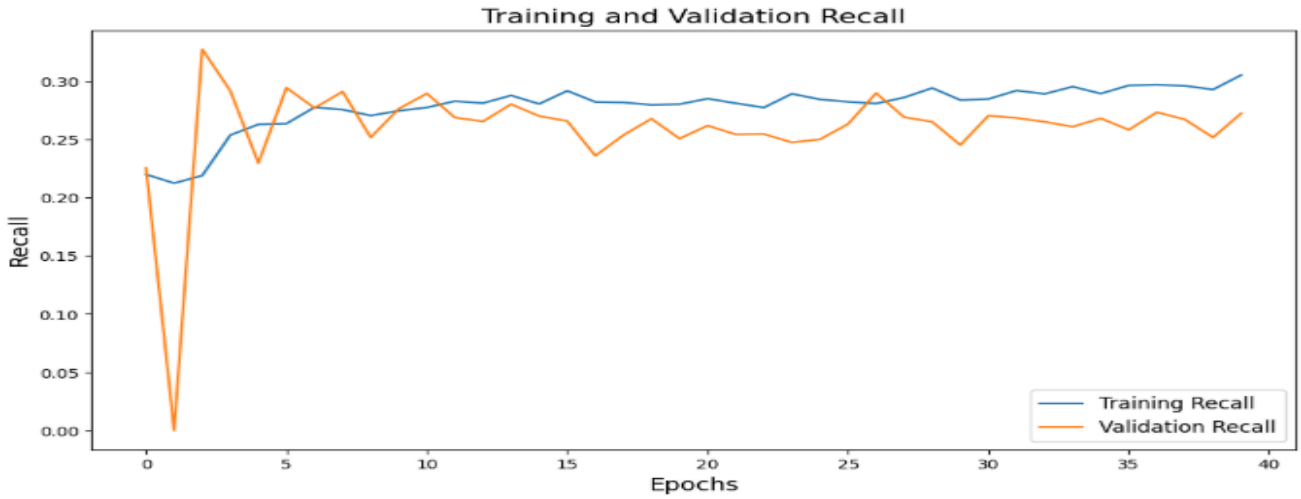
- The training and validation precision both start low and gradually increase over the epochs.
- The training precision is higher than the validation precision throughout, indicating some overfitting.
- Both precision curves stabilize after around 25 epochs, with the validation precision reaching around 0.65.



Training and Validation Recall:

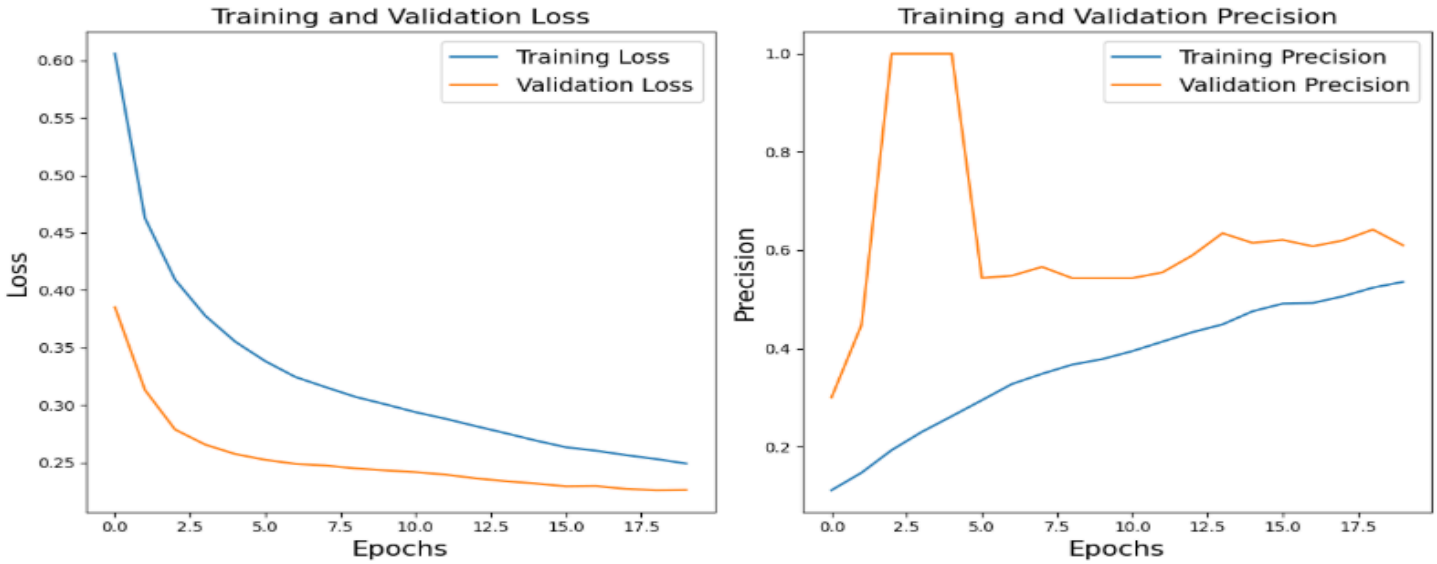
- The training recall starts high, drops sharply, then fluctuates before stabilizing.
- The validation recall starts lower, increases gradually, and also stabilizes after 25 epochs.
- The training and validation recall curves are closer together compared to the precision, indicating the model is better at recalling positive instances.

Overall, the model appears to be converging and stabilizing after around 25 epochs, with the training performance slightly better than the validation performance, suggesting some overfitting. The recall metrics are stronger than the precision, indicating the model may be better at identifying positive instances than accurately classifying .



INSIGHTS FROM LSTM:

The graphs in the images show the training and validation metrics for a machine learning model, likely a recurrent neural network like an LSTM, over the course of training.



Looking at the Loss and Precision graphs in Image 1:

- The training loss starts very high and rapidly decreases in the first few epochs, while the validation loss decreases more gradually.
- The gap between the training and validation loss narrows over time, indicating the model is generalizing better.
- Both the training and validation precision start low but increase steadily, with the training precision staying higher than the validation precision.

Examining the Recall graph in Image 2:

The training recall starts high, drops sharply, then fluctuates before stabilizing.
The validation recall starts lower, increases gradually, and also stabilizes after around 15 epochs.
The training and validation recall curves are closer together compared to the precision, indicating the model is better at recalling positive instances than accurately classifying them.

Overall, the model appears to be converging and stabilizing after around 15-20 epochs. The training performance is slightly better than the validation performance, suggesting some overfitting. However, the model seems to have a stronger ability to recall positive instances than to precisely classify them.These insights can help the model developers understand the strengths and weaknesses of the model, and guide them in making further improvements, such as adjusting hyperparameters, adding regularization, or collecting more diverse training data to address the overfitting and precision issues.

Report on Model Classification Performance

To evaluate the performance of the CNN and LSTM models, I selected three example films from the dataset and generated predictions for their genres.For each film, I first plotted the movie poster and printed the overview text. I then used the pre-trained CNN model to generate the top 3 most probable genre predictions based on the poster image, and the LSTM model to generate the top 3 genre predictions based on the overview text. I compared these predicted genres to the ground truth genres for each film.

Example 1:
- Film Overview: "Follows hard-to-please Aurora looking for love, and her daughter's family problems."
- Ground Truth Genres: ['Comedy', 'Drama']
- CNN Top 3 Predicted Genres: ['Family', 'Sci-Fi', 'Documentary']
- LSTM Top 3 Predicted Genres: ['Family', 'Crime', 'Adventure']

Example 2:
- Film Overview: "A Pittsburgh woman with two jobs as a welder and an exotic dancer wants to get into ballet school."
- Ground Truth Genres: ['Drama', 'Music', 'Romance']
- CNN Top 3 Predicted Genres: ['Family', 'War', 'Adventure']
- LSTM Top 3 Predicted Genres: ['Family', 'Crime', 'Sci-Fi']

Example 3:
- Film Overview: "A snobbish investor and a wily street con artist find their positions reversed as part of a bet by two callous millionaires."
- Ground Truth Genres: ['Comedy']
- CNN Top 3 Predicted Genres: ['Family', 'Crime', 'Adventure']
- LSTM Top 3 Predicted Genres: ['Family', 'Crime', 'Sci-Fi']

Analysis:
The performance of the CNN and LSTM models on these example films is mixed. For the first film, neither model was able to accurately predict the ground truth genres of Comedy and Drama. Instead, the models predicted genres like Family, Sci-Fi, Crime, and Adventure.For the second film, the models again struggled, with the top predictions being quite different from the ground truth genres of Drama, Music, and Romance. The CNN model predicted genres like Family, War, and Adventure, while the LSTM model predicted Family, Crime, and Sci-Fi.The third film, a Comedy, was also not properly classified by either model. Both the CNN and LSTM models predicted genres like Family, Crime, and Adventure, missing the ground truth Comedy genre.

CONCLUSION:
Overall, the results suggest that both the CNN and LSTM models still have difficulty accurately predicting the full set of genres for these example films. The models seem to have a bias towards predicting more general genres like Family, Crime, and Adventure, rather than the more specific and nuanced genres present in the ground truth. Further work may be needed to improve the models' multi-label classification capabilities and their ability to capture the full breadth of genre representation in the dataset.